Grip or glide? The surface challenge · Knowledge Organiser

Learning goal: I can explain how friction acts between touching surfaces and judge when grip or glide is useful.

Today I Can

I can identify the two touching surfaces where friction acts.

I can draw a friction arrow opposite the direction of sliding.

I can explain one job where friction helps and one where it is unhelpful.

Must-Know Words

friction: a contact force between two surfaces that slide or try to slide past each other

contact force: a force that acts only when two objects or surfaces touch

grip: high resistance to slipping between two touching surfaces

lubrication: adding oil or grease to reduce unwanted friction between moving parts

Core Science Steps

Friction is a contact force between two surfaces that slide, or try to slide, past each other.

The force acts opposite to the relative sliding and

depends on both materials in the surface pair.

Grip is helpful for brakes and shoe soles, whereas unwanted rubbing heats and wears machine parts.

Keep the Science Clear

Naming both touching surfaces and the direction of sliding makes every force explanation clear.

Looking at one surface alone is not enough·friction depends on the pair of surfaces touching.